INCOMPLETE WORK SUMMARY

Incomplete Work and Prioritization Rationale

Overview

The assessment included a broad range of advanced requirements covering data engineering, exploratory analytics, statistical analysis, machine learning, and innovation-focused solution design.

Due to time constraints, a prioritization strategy was adopted to maximize the quality and completeness of the most business-critical components.

Statistical Analysis

Several advanced statistical analysis activities were not completed in full.

These included:

- Formal hypothesis testing
- Confidence interval estimation
- Statistical significance testing
- Regression modeling
- Effect size analysis

While these activities would have strengthened analytical validation, the exploratory findings already provided substantial business insight.

Machine Learning Components

Predictive machine learning models were not implemented.

Potential future models include:

- Price prediction
- Demand forecasting
- Occupancy prediction
- Revenue prediction

The decision was made to prioritize data engineering and business analysis rather than model development.

Open Innovation Challenge

The assessment included an open innovation component requiring the development of an advanced business solution.

Although preliminary ideas were explored, a complete architecture, prototype, and evaluation were not developed within the available timeframe.

Interactive Business Intelligence Dashboard

The project did not include the implementation of Power BI or Tableau dashboards.

Instead, visualization efforts focused on notebook-based analytical reporting.

Prioritization Strategy

The decision to prioritize engineering and exploratory analysis was based on three factors:

1. Demonstrating core Data Engineering competencies.
2. Delivering meaningful business insights.
3. Producing a professional and complete analytical report.

This strategy ensured that available time was invested in areas most relevant to the target internship role.

Conclusion

Although some advanced components remain incomplete, the project successfully delivered a substantial analytical solution demonstrating practical data engineering, data preparation, feature engineering, visualization, and business intelligence capabilities.